

ex 20

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```
1 clear;
2 clc;
3 close all;
4
5 %% 4x4 MIMO Planar Array - 3D Beam Steering
6 % No Toolbox Required
7
8 N = 4;
9 d = 0.5;
10
11 % Desired beam direction
12 theta0 = 30*pi/180;
13 phi0 = 45*pi/180;
14
15 %% Angular grid
16
17 theta = linspace(0,pi,91);
18 phi = linspace(0,2*pi,181);
19
20 [TH,PH] = meshgrid(theta,phi);
21
22 %% Calculate Array Factor
23
24 AF = zeros(size(TH));
25
26 for m = 0:N-1
27     for n = 0:N-1
28         psi = 2*pi*d*( ...
29             m*(sin(TH).*cos(PH) - sin(theta0)*cos(phi0)) ...
30             + n*(sin(TH).*sin(PH) - sin(theta0)*sin(phi0)) );
31
32         AF = AF + exp(1i*psi);
33
34     end
35 end
36
37
```

Run

Save

Output

Input

4x4 MIMO BEAM STEERING SIMULATION

Number of Elements : 4 x 4
Element Spacing : 0.50 wavelength
Elevation Angle : 30.0 degrees
Azimuth Angle : 45.0 degrees

3D beam pattern generated successfully.

[Execution complete with exit code 0]

